

Time: 2hrs	School Of Engineering, CUSAT First Internal Examination of S-3B.Tech Mechanical Engineering, January 2023 19-205-0305 Metallurgy and Material Science	Marks:50
------------	--	----------

	CO	BL	Mark
--	----	----	------

Part-A

1	How can we classify the structure of materials based on the magnification and resolution (level) of observation? State the levels of structure which are of greatest interest in material science and engineering?	CO1	L1	2
2	Differentiate between stability and metastability. A metastable state of existence is very common in materials. Give an example.	CO1	L2	2
3	There is no end-centred tetragonal lattice in the Bravais list, but there is an end-centred orthorhombic lattice. Explain why this is so.	CO1	L4	2
4	Draw the following directions and planes in cubic crystal: (a) [011] (b) [112] (c) (111) (d) (112).	CO1	L3	2
5	What is the difference between homogenous and heterogeneous nucleation.	CO1	L2	2
6	State and explain the importance of Gibbs phase rule.	CO2	L2	2
7	Write note on different types of solid solution	CO2	L2	2

Part-B

8	Explain the concepts of space lattice, basis and crystal structure. How are they related?	CO1	L2	4
9	Compute the Atomic Packing Factor (APF) for Body Centred Cubic (BCC) and Face Centred Cubic (FCC) crystal structures.	CO1	L2	4
10	Define planar atomic density on a crystal plane and linear atomic density in a direction in crystal structures. Why they are important?	CO1	L2	4
11	Copper has an atomic radius of 0.128 nm, an FCC crystal structure, and an atomic weight of 63.5 g/mol. Compute its theoretical density.	CO1	L3	4
12	Describe the crystalline imperfections based on their geometry.	CO1	L2	4
13	Differentiate between edge dislocations and screw dislocations.	CO1	L2	4
14	State and explain Hume – Rothery's rules	CO2	L2	4
15	Discuss with neat diagram on allotropic forms of iron.	CO2	L3	4
16	Explain Cu- Ni binary alloy system with neat figure.	CO2	L3	4